

29 Beyond Perturbation Theory

Most of the implications of supersymmetry discussed so far have been inferred with the use of perturbation theory. In this chapter we shall consider some results that apply even when non-perturbative effects are taken into account.

29.1 General Aspects of Supersymmetry Breaking

Supersymmetry is not observed in the spectrum of known particles, so it must be broken. We saw in the previous chapter that supersymmetry breaking in the tree approximation of the standard model is experimentally ruled out, and that the large disparity between the electroweak breaking scale and the Planck or grand unification scales suggests that supersymmetry is likely to be broken when some running gauge coupling becomes strong. It is therefore essential for us to explore spontaneous supersymmetry breaking without using perturbation theory.

We saw in Section 26.7 that the supersymmetry of the action implies the existence of a supersymmetry current $S^\mu(x)$. This current is a Majorana spinor in the sense of Eq. (26.A.2):

$$S^\mu(x)^* = -\beta\gamma_5\epsilon S^\mu(x); \quad (29.1.1)$$

it is conserved,

$$\partial_\mu S^\mu_\beta(x) = 0; \quad (29.1.2)$$

and the integral of its time component is the supersymmetry generator

$$\int d^3x S^0_\beta(x) = Q_\beta, \quad (29.1.3)$$

for which the commutator of $-i(\bar{\alpha}Q)$ with any operator gives the change of that operator under a supersymmetry transformation with infinitesimal Majorana spinor parameter α .

The arguments that led to these results relied on the supersymmetry of the action; nothing depended on whether or not supersymmetry is spontaneously broken, except perhaps for the assumption that the integral (29.1.3) exists. In fact this assumption may be violated in theories with massless fermions, which can have long range effects (or, equivalently, poles at zero four-momentum) that would make this integral not converge. We will see here that such massless fermions are necessary consequences of supersymmetry breaking. In order to avoid the issue of the convergence of this integral even in theories with massless fermions, it is very convenient to work in a space of finite volume V . We can do this while maintaining translation invariance by imposing periodic boundary conditions: all fields are assumed to be unaffected by translations of any spatial coordinate x^i by an amount $V^{1/3}$.

The existence of an operator Q_α that induces supersymmetric transformations on quantum fields would allow us to derive all the consequences of supersymmetry, provided that there exists a supersymmetric vacuum state $|\Omega\rangle$ of zero three-momentum from which multiparticle states may be constructed by acting with field operators. But if $|\Omega\rangle$ is supersymmetric in the sense that $Q_\alpha|\Omega\rangle = 0$, then it follows from the anticommutation relation

(25.2.36) that this state has zero energy as well as zero momentum. Conversely, by taking the vacuum expectation value of the positive operator (returning for a moment to two-component notation) $\{Q_a, Q_a^*\}$, we see that if the vacuum has zero energy then it must be annihilated by Q_a and Q_a^* , and hence be supersymmetric, while if it is not supersymmetric then its energy must be positive-definite. *The question of whether supersymmetry is or is not spontaneously broken is thus entirely a question of whether the vacuum has a positive-definite or a zero energy.*

The same reasoning led Witten to the conclusion that extended supersymmetry with $N > 1$ two-component spinor generators Q_{ar} and their adjoints cannot be spontaneously broken to extended supersymmetry with fewer generators or to simple supersymmetry, because if any one of the generators does not annihilate the vacuum then the vacuum energy cannot vanish, and it follows from this that none of the generators can annihilate the vacuum [1]. Usually the energy of the vacuum state appears as an ill-defined additive constant in the energies of all states, but here this constant is given a meaning by the appearance of the energy-momentum four-vector in the anticommutation relations of supersymmetry. One of the advantages of working in a finite volume is that it makes it meaningful to talk about the total energy of the vacuum.

Hughes, Liu, and Polchinski have pointed out that there are theories that exhibit a sort of partly broken supersymmetry [1a]. These theories do not have a supersymmetry algebra of the sort described in Chapter 25. Instead, they have an algebra of currents, based on anticommutation relations like Eq. (26.7.45):

$$\int d^3x \left\{ S_{r\alpha}^0(x), \bar{S}_{s\beta}^\mu(y) \right\} = -2i\delta_{rs}\gamma_\nu\Theta^{\mu\nu}(y) + 2i\gamma_{\alpha\beta}^\mu C_{rs},$$

where $\Theta^{\mu\nu}$ is an energy-momentum tensor satisfying the conservation condition $\partial_\mu\Theta^{\mu\nu} = 0$, and C_{rs} is a new ingredient, a constant. For $N = 1$ this constant could be regarded as a term $-\eta^{\mu\nu}C$ in $\Theta^{\mu\nu}$, but this is not possible for extended supersymmetry unless $C_{rs} \propto \delta_{rs}$, which need not be the case. This algebra is not ruled out by the Haag–Lopuszanski–Sohnius theorem proved in Section 25.2 because it could not be a symmetry of the S -matrix. It is true that spontaneously broken symmetries are never symmetries of the S -matrix, but they are usually assumed to be based on algebras or superalgebras that *could* be symmetries of the S -matrix in some phases of some theories. With C_{rs} not proportional to δ_{rs} , the superalgebra of currents is one that could not generate a symmetry of the S -matrix in any phase of any theory. Here we will only consider superalgebras of the sort described in Chapter 25, to which Witten’s argument does apply.

Another advantage of working in a finite volume is that all states become discrete and normalizable. An immediate consequence of the fact that Q_a commutes with P_μ is that any state of non-zero energy is paired with another state of the same energy and momentum but opposite statistics. To see this, note that for any three-momentum $\mathbf{p}$ we may find a two-component spinor u_a such that $\sum_{ab} u_a^* \sigma_{ab} \cdot \mathbf{p} u_b = 0$ and $\sum_a |u_a|^2 = 1$. (For $\mathbf{p}$ in the 3-direction, take $u = (1, 1)/\sqrt{2}$. For $\mathbf{p}$ in any other direction, apply to this u the spin 1/2 representation of the rotation that takes the 3-direction into the direction of $\mathbf{p}$.) Then, within the space with four-momentum p^μ , the anticommutation relations (25.2.31)

and (25.2.32) give

$$Q^2(p) = p^0, \quad (29.1.4)$$

where $Q(p)$ is the Hermitian linear combination of supersymmetry generators:

$$Q(p) = \sum_a u_a Q_a + \sum_a u_a^* Q_a^*. \quad (29.1.5)$$

Acting with $Q(p)$ on any normalized state $\|X\rangle$ with four-momentum p^μ and $p^0 > 0$, we get another normalized state $\|Y\rangle = Q(p)\|X\rangle/\sqrt{p^0}$ with opposite statistics and the same four-momentum. Furthermore, $\|X\rangle$ is the only state related to $\|Y\rangle$ in this way, because if $\|Y\rangle = Q(p)\|X\rangle/\sqrt{p^0}$ then, according to Eq. (29.1.4), $\|X\rangle = Q(p)\|Y\rangle/\sqrt{p^0}$. The multiplicity of supersymmetry generators and of spin states will usually cause these pairs of fermionic and bosonic states to be joined by other pairs, all with the same four-momentum, but for the moment it is enough to know that all states of non-zero energy may at least be grouped in these pairs.

When supersymmetry is broken we do not expect states with a definite number of particles to form supermultiplets with other states of opposite statistics and with the same four-momentum and the same number of particles. The pairing of states then requires the existence of a massless fermion, so that an n -particle state can be paired with a state of the same energy and momentum but opposite statistics, consisting of the same n particles together with a massless fermion of zero energy and momentum. This massless fermion is known as a *goldstino*. To be more precise, any n -particle state is accompanied with two states of the same energy and momentum and opposite statistics, containing an additional zero-momentum goldstino of spin up or down, and with another state of the same energy and momentum and the *same* statistics, containing two additional zero-momentum goldstinos of opposite spin. In particular, when supersymmetry is spontaneously broken the vacuum state has non-zero energy, so it must be paired with a fermionic state of the same energy and zero momentum; more precisely, the vacuum and the state containing two zero-momentum goldstinos are paired with the two states of a single zero-momentum goldstino. It is only when supersymmetry is unbroken that there is a state of zero energy, the vacuum, that can be unpaired.

The pairing of states of non-zero energy provides a valuable diagnostic tool that can in some cases tell us that supersymmetry is not spontaneously broken, even where perturbation theory is not adequate to answer this question. When all interactions are weak, we can rely on perturbation theory to give us a *qualitative* picture of the spectrum. If it turns out that in the tree approximation there are n vacuum states with zero energy, and no massless fermions, then we can be confident that for weak coupling there are no zero-energy fermionic states with which we could pair the n vacuum states, so these unpaired states would have to have precisely zero energy. Then, as we increase the strength of the couplings or vary the parameters of the theory in any other way, states may move from positive to zero energy or vice-versa, but they will generally not suddenly appear or disappear. (There is an exception for changes in the parameters that change the asymptotic behavior of the Lagrangian for large fields; as we will shortly see, this can produce or destroy states.) Because each state of non-zero energy is always paired with another of opposite statistics, they can only make the

transition from zero to non-zero energy or vice-versa in such pairs, so the number of bosonic zero-energy states minus the number of fermionic zero-energy states does not change as the parameters of the theory are varied, as long as the large-field behavior of the Lagrangian is not changed. This difference is known as the *Witten index* [2]. Formally, this index is $\text{Tr}(-1)^F$, where F is the fermion number; the pairing of states discussed earlier insures that this trace can receive no contribution from states of non-zero energy. If the Witten index is non-zero then there must be *some* states of zero energy, and so supersymmetry cannot be broken. In particular, in a theory where the tree approximation gives n zero-energy vacuum states and no zero-energy fermions, the Witten index is n for weak coupling, where the tree approximation can be trusted to give a qualitative picture of the spectrum, and the index remains equal to n as the strength of the coupling is increased, so we can be certain that higher-order effects or even non-perturbative effects do not break supersymmetry.

As an example of the use of the Witten index, consider the Wess–Zumino theory of a single chiral superfield with a cubic polynomial superpotential of the form (26.4.16):

$$f(\phi) = \frac{1}{2}m^2\phi^2 + \frac{1}{6}g\phi^3,$$

where ϕ is the complex scalar field component of the superfield. We saw in Section 26.4 that this model does not exhibit supersymmetry breaking in the tree approximation, but what about higher orders of perturbation theory, and what about non-perturbative effects? Perturbation theory gives a good approximation to the energy spectrum when m is large and g is small; it tells us that in this case there are two bosonic states near zero energy, corresponding to the solutions $\phi = 0$ and $\phi = -2m^2/g$ of the equation $\partial f(\phi)/\partial\phi = 0$, and no fermionic states near zero energy; the lowest-energy fermionic state is a zero-momentum one-fermion state of energy near $|m|$. In typical scalar field theories we would not expect the two bosonic states to have precisely zero energy; even though each has zero energy in the tree approximation, higher-order effects (including tunneling through the barrier between $\phi = 0$ and $\phi = -2m^2/g$) would be expected to mix them and shift their energies away from zero. (It is only in the limit of infinite volume that this barrier becomes impassable.) But in supersymmetric theories these states must have precisely zero energy, because there is no low-energy fermion state with which they could be paired. Thus for m large and g small the Witten index is 2. Because the Witten index is invariant under changes in the parameters of the theory, the Witten index remains equal to 2 even when g is large, where perturbation theory breaks down, and even when m vanishes, where the two potential wells merge. (It is not easy to calculate the Witten index directly in this case, because of the presence in the tree approximation of massless bosons as well as massless fermions.) Since the Witten index is not zero, supersymmetry remains strictly unbroken in the Wess–Zumino model, whatever the values of its parameters.

The same arguments can generally be used in theories with several chiral scalar superfields to show that the Witten index is positive, and that supersymmetry is therefore not spontaneously broken. The O’Raifeartaigh models discussed in Section 26.5 are an exception, because there are flat directions in which the potential remains constant as the fields go to infinity, rather than growing as some power of the fields. These models provide a good

illustration of the fact that although the Witten index must be zero for supersymmetry to be broken, a zero Witten index does not necessarily imply that supersymmetry is broken. For instance, if we write the superpotential used as an illustration in Section 26.5 in terms of canonically normalized superfields, then it takes the form

$$f(X, Y_1, Y_2) = mY_1(X - a) + gY_2X^2,$$

with arbitrary parameters m , g , and a . The potential is then

$$U(x, y_1, y_2) = |g|^2|x|^4 + |m|^2|x - a|^2 + |my_1 + 2gxy_2|^2,$$

with lower-case letters denoting the scalar components of the left-chiral superfields. For m and a non-zero and g small, perturbation theory gives a good estimate of the spectrum, and tells us that there is a minimum of the potential near $x = a - 2|g|^2|a|^4/|m|^2$ with $my_1 + 2xy_2 = 0$, at which the vacuum energy is approximately equal to $|ga^2|^2V$. Since this energy appears as an additive constant in all states, there are no zero-energy states, and the Witten index vanishes. The matrix $\mathcal{M}$ of second derivatives of the superpotential (with rows and columns labelled in the order x, y_1, y_2) is here

$$\mathcal{M} = \begin{pmatrix} 2gy_2 & m & 2gx \\ m & 0 & 0 \\ 2gx & 0 & 0 \end{pmatrix}.$$

This has an eigenvector $(0, 2gx, -m)$ with eigenvalue zero, so there is a massless fermion here; this is the goldstino associated with the breaking of supersymmetry. The fermionic state degenerate with the vacuum consists of a single goldstino of zero energy and momentum. (Once again, there are two vacuum-energy fermionic states, with opposite orientations for the goldstino spin, and two vacuum-energy bosonic states: the vacuum, and a state containing two goldstinos with opposite spin.) Now, as $a \rightarrow 0$ supersymmetry may become unbroken (and we shall see later that it does), but the Witten index must remain 0; in this case the massless fermion is no longer a goldstino, but continuity demands that its mass remains zero, so it remains paired with the vacuum state. This of course is a general feature of theories in which supersymmetry is restored at isolated values of the parameters; continuity demands that the massless fermion that plays the role of the goldstino when supersymmetry is broken remains massless (though no longer a goldstino) at the value of the parameters where supersymmetry is restored, so that the vacuum remains paired with the massless fermion state, and the Witten index remains zero.

This model provides a good illustration of why we had to qualify the statement that the Witten index does not change when we vary the parameters of a supersymmetric theory, with the proviso that the parameters must not be changed in such a way as to alter the asymptotic behavior of the Lagrangian density for large fields. Suppose we unflatten the flat direction in this model by adding a small term to the superpotential, so that it now reads

$$f(X, Y_1, Y_2) = mY_1(X - a) + gY_2X^2 + \frac{1}{2}\epsilon(Y_1^2 + Y_2^2),$$

where ϵ is a small mass parameter. Now there are two solutions of the conditions for supersymmetry to be preserved:

$$0 = \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y_1} = \frac{\partial f}{\partial y_2}.$$

At these solutions, x is at one of the roots of the quadratic equation $2g^2x^2 + m(x - a) = 0$, while y_1 and y_2 are of order $1/\epsilon$: $y_1 = -m(x - a)/\epsilon$ and $y_2 = -gx^2/\epsilon$. We see that the reason that the Witten index can change from 0 to 2 when we turn on the small parameter ϵ is that two new minima of the potential come in from infinite field values.

In deciding whether or not supersymmetry is broken in theories with a vanishing Witten index, it is often useful to use conservation laws to limit the pairings that may occur, and to define a new sort of index. If K is a quantum operator that commutes with the supersymmetry generators Q_α (and hence also with the Hamiltonian), then all states of non-zero energy that have a definite value for K are paired with states of opposite statistics and the same energy and momentum *and* the same value of K . Also, not only is the Witten index $\text{Tr}(-1)^F$ independent of the parameters of the theory (as long as they are not varied in a way that varies the large-field asymptotic behavior of the Lagrangian)—so also is the *weighted Witten index*, given by $\text{Tr} g(K)(-1)^F$, where $g(K)$ is an arbitrary function of the conserved quantity. To use the conservation law in this way, it is not necessary that it be unbroken when the volume V becomes infinite; it is only necessary that K commute with the supersymmetry generators.

In diagnosing the possibilities of supersymmetry breaking, it is sometimes helpful to work with a linear combination of weighted Witten indices for a number of different conserved quantities. In particular, consider the quantity

$$W_G = \sum_{h \in G} \text{Tr} \{ h(-1)^F \} , \quad (29.1.6)$$

with the sum running over all elements of some symmetry group G . (For compact continuous groups, this sum should be interpreted as an integral over the group volume, with a suitable invariant measure.) Within any irreducible representation other than the identity, the ‘characters’ $\text{Tr} h$ add up to zero when summed over a finite or compact group, so

$$W_G = \sum_f N(f)(-1)^f , \quad (29.1.7)$$

where $N(f)$ is the number of times that the identity representation of G appears among states with fermion number f . In other words, W_G is just the Witten index, but evaluated using only G -invariant states. As long as G is conserved, W_G remains independent of the parameters of the theory, and if non-zero indicates that supersymmetry is unbroken.

Conservation laws will be used in this way in Section 29.4 to study the spontaneous breaking of supersymmetry in gauge theories, but a simpler (though academic) example is provided by the O’Raifeartaigh-style model discussed earlier, only now with the parameter a set equal to zero. The superpotential here is

$$f(X, Y_1, Y_2) = mY_1X + gY_2X^2 , \quad (29.1.8)$$

yielding a tree-approximation potential

$$U(x, y_1, y_2) = |m|^2|x|^2 + |g|^2|x|^4 + |my_1 + 2gxy_2|^2. \quad (29.1.9)$$

We know that the Witten index vanishes here, because we have seen that it vanished for $a \neq 0$, but is supersymmetry broken? With $a = 0$, there are now field values with $x = y_1 = 0$ where the potential vanishes in the tree approximation, but how can we tell whether effects of higher order in g or even non-perturbative effects give the corresponding states a small energy? To answer this, we note that this superpotential (and hence the Lagrangian density) is invariant under a discrete symmetry K , under which the superfields are transformed by

$$KXK^{-1} = iX, \quad KY_1K^{-1} = -iY_1, \quad KY_2K^{-1} = -Y_2. \quad (29.1.10)$$

(Note that this symmetry is violated by the term $-maY_1$ in the original superpotential, so none of the results obtained using K will apply to that superpotential.) Since the potential vanishes for $x = y_1 = 0$ and y_2 arbitrary, for small g we may use perturbation theory to tell us that for each y_2 there is a bosonic vacuum state *near* zero energy. For $y_2 = 0$ this vacuum is even under K . For any non-zero value of $|y_2|$ we can take linear combinations of the two zero-energy states with $y_2 = \pm|y_2|$, of which one will be even under K and the other odd. As we have seen, there is also a massless fermion here, the fermionic component of Y_2 , but this fermion is odd under K , so it cannot be paired with the even vacuum states. The only other fermions in the theory have tree-approximation masses $|m|$, so for small g they can't be paired with the even vacuum states either. We conclude then that for small g the even vacuum states must have precisely zero energy, and supersymmetry is unbroken. It is not easy to calculate the weighted Witten index here, because there are an infinite number of bosonic zero-energy states that are even under K , consisting of either zero or two Y_2 -fermions with zero momentum plus any even number of zero-momentum Y_2 bosons, but it is clear that $\text{Tr } K(-)^F > 0$, and since this is independent of g (as long as $g \neq 0$), supersymmetry cannot be broken for any finite g .